Effect of Honeycomb Geometry on Additively Manufactured Kevlar-Reinforced Ballistic Armor

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The increasing availability of fiber-reinforced additive manufacturing has created opportunities for developing lightweight structures capable of improving energy absorption in ballistic armor systems. This study investigated the incorporation of continuous Kevlar fiber-reinforced, additively manufactured components into a multilayer armor system, with emphasis on how cellular geometry influenced impact response and load distribution. Previous testing of individual honeycomb structures demonstrated insufficient penetration resistance when used without a protective front layer, creating a need to evaluate these structures as energy-absorbing backing components.

Test specimens were manufactured on a Markforged Mark Two printer using Onyx thermoplastic reinforced with continuous Kevlar fiber. Three honeycomb-derived geometries were evaluated: a conventional honeycomb, a two-sided offset honeycomb in which cell centers on one side aligned with cell corners on the opposite side, and a three-dimensional cellular structure containing a staggered arrangement of elongated hexagonal bipyramidal cavities. Specimens maintained consistent unit-cell size, overall dimensions, and solid infill, although these constraints resulted in varying plastic and fiber volumes among geometries. Each printed structure was positioned behind a sewn Kevlar panel and subjected to 9-millimeter hollow-point ammunition fired from 5 m. The sewn layer was intended to resist penetration, while the printed backing structure was designed to deform, absorb impact energy, limit panel deflection, and distribute the resulting load.

Ballistic testing remained in progress, and definitive quantitative results had not yet been established. The study hypothesized that the modified cellular geometries would produce a more uniform response across varied impact locations while improving outward load distribution. The resulting data will be used to compare the effectiveness of each geometry as an energy-absorbing backing structure and to assess the potential of additively manufactured Kevlar-reinforced components for lightweight, customizable multilayer ballistic armor. Future work may also evaluate polyurethane or nanomaterial interlayers to further improve energy absorption.